CBCS SCHEME

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BPLCK105B/BPLCKB105

First Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Introduction to Python Programming

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	List and explain the use of comparison operator in python. Write the step by step execution of the following expression in python. 3/2 * 4 + 3 + (10/4)**3 – 2	6	L1	CO1
	b.	Explain the control statements, if, else, elif with proper syntax and examples.	6	L2	CO1
	c.	Develop a python program to calculate the area and circumference of a circle input the value of radius and print the results.	8	L3	CO1
		OR			
Q.2	a.	Explain the string concatenation and string replication operator with an example.	6	L2	CO1
	b.	Explain local and global scope of variable with suitable example.	6	L2	CO1
	c.	Develop a program to read the student details Like Name, USN and Marks in three subjects. Display the student details, total marks and percentage with suitable messages.	8	L3	CO1
		Module – 2			
Q.3	a.	What is list? Explain the concept of list indexing and slicing with examples.	6	L2	CO2
	b.	With suitable examples, explain the list methods append( ), extend( ), sort( ), count( ) and pop( ).	8	L2	CO2
	c.	Read N numbers from the console and create a list. Develop a program to print mean, variance and standard deviation with suitable message.	6	L3	CO2
		OR			
Q.4	a.	Define tuple data type? List out the difference between tuple and list.	6	L2	CO2
	b.	Identify and explain the dictionary methods like get( ), item( ), keys( ) and values ( ) in python with examples.	8	L2	CO2
	c.	Develop a python program to swap two numbers without using Intermediate variables. Prompt the user for input.	6	L2	CO3

Module – 3					
Q.5	**a.**	Write the output of the following : i) 'HeLLo' · upper( ) · isupper ( ) ii) 'HeLLo' · upper( ) · lower( ) iii) '__' · Join('There can be only one' · split( ))	**6**	**L2**	**CO3**
	b.	With examples, explain any five string methods.	**6**	**L2**	**CO3**
	c.	Develop a python program to count the total number of vowels, consonants in a string.	**8**	**L3**	**CO3**
OR					
Q.6	**a.**	Make use of the concept of file handling and explain Reading and writing process with suitable python programs.	**7**	**L2**	**CO3**
	b.	Explain the concept of file path, also discuss absolute and relative paths.	**7**	**L2**	**CO3**
	c.	Briefly, explain saving variables with shelve module.	**6**	**L2**	**CO3**
Module – 4					
Q.7	**a.**	Explain the following file operations in python with suitable examples. i) Copying files and folders ii) Moving files and folders iii) Permanently deleting files and folders	**6**	**L2**	**CO3**
	b.	List out the benefits of compressing file with zip file module, also explain the concepts of walking a directory tree.	**8**	**L2**	**CO3**
	c.	List out the difference between shutil.copy( ) and shutil.copytree( ) methods.	**6**	**L3**	**CO3**
OR					
Q.8	**a.**	Briefly explain Assertion and raising a exception.	**6**	**L2**	**CO3**
	b.	Develop a python program with a function named DivExP which takes two parameters a, b and returns a value C. (C = a/b). Write suitable assertion for a > 0 in function DivExP and raise on exception for when b = 0. Program has to read two values from the console and call a function DivExP.	**8**	**L3**	**CO3**
	c.	Briefly explain the difference logging levels.	**6**	**L2**	**CO3**
Module – 5					
Q.9	**a.**	Define classes and objects in Python. Construct the class called rectangle and initialize it with height = 100, width= 200, starting point as (x = 0, y = 0) and write the method to display the center point coordinates of a rectangle.	**8**	**L2**	**CO4**
	b.	Briefly explain the concept of prototyping and planning.	**6**	**L2**	**CO4**
	c.	Explain Clearly _ _init( )_ _ and _ _str( )_ _ method with examples.	**6**	**L2**	**CO4**
OR					
Q.10	**a.**	Explain the term objects are mutable with an example.	**6**	**L2**	**CO4**
	b.	Explain the concept of polymorphism with examples.	**8**	**L2**	**CO4**
	c.	Explain briefly pure functions and modifiers with examples.	**6**	**L2**	**CO4**

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Scheme & Solution

Signature of Scrutinizer

Subject Title: Introduction to Python Programming Subject Code: BPLCK105B

Question Number	Solution	Marks Allocated
1	MODULE - I	
(a)	== Equal to, != Not equal to < Less than, > Greater than <= >= Briefly Explain	03
	$3/2 * 4 + 3 + \left(\frac{10}{4}\right) ** 3 - 2$ $3/2 * 4 + 3 + (2.5) ** 3 - 2$ $6 + 3 + 15.625 - 2$ $\boxed{= 22.625}$	03
(b)	if, elif and else is a multiway decision control statement - Explain briefly	02
	Eg:- if Boolean_Expression_1 : stmt_1	02
	elif Boolean_Expression_2 : stmt_2 ⋮	02
	else : Final_stmt	
(c)	radius = int(input('Enter radius of circle'))	02
	area_of_circle = 3.14 x radius x radius	02
	cir_of_circle = 2 x 3.14 x radius	02
	Print (area_of_circle) Print (cir_of_circle)	02

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Question Number	Solution	Marks Allocated
2 (a)	String concatenation - + is used to join two strings.	03
	Eg :- 'one' + 'two' 'onetwo'	
	String Replication - * operator on one string and one integer value.	03
	Eg 'one' * 2 'one one'	
(b)	variables defined outside all functions is Global.	02
	Variables defined in a function are said to exist in that function are LOCAL	02
	Eg :- x = 0 — Global def function1( ): y = 0 — Local	02
(c)	name = input('Enter student name') usn = input('Enter USN') sub1 = int(input('Enter marks scored in sub1')) sub2 = int(input('Enter marks scored in sub2')) sub3 = int(input('Enter marks scored in sub3')) total = sub1 + sub2 + sub3	02

	Percentage = ~~sub3~~. total/3
	 if (Percentage >= 70 and sub1 > 35 and sub2 > 35 and sub3 > 35):
	 grade = 'First class with distinction'
	 elif (Percentage < 70 and Percentage > 60 and sub1 > 35 and sub2 > 35 + sub3 > 35):
		02

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Question Number	Solution	Marks Allocated
	grade = 'First class'	
	elif (Percentage < 60 and Percentage >= 35 and sub1 > 35 and sub2 > 35 and sub3 > 35) :	02
	grade = 'Second Class'	
	else: grade = 'Fail'	
	print('Name:', name)	
	print('USN:', usn)	
	Print('subject1 Marks', sub1) Print('subject2 mark', sub2)	
	print('subject3 mark', sub3)	02
	Print('Total marks', total) Print('%', Percentage)	
	print('your Grade is', grade)	
	MODULE - II	
3 (a)	A list is a value that contains multiple values in an ordered sequence.	02
	Eg :- ['cat', 'bat', 'rat']	
	Indexing - spam = ['cat', 'bat', 'rat']	02
	spam[0] = 'cat'	
	slicing - spam [0 : 3] = ['cat', 'bat', 'rat']	02
(b)	a = [1, 2, 3, 4, 5]	
	a.append(6) - Adds a single item to the end of list	02
	extend() - Adds items in list2 to the end of list	02
	sort() - sorts items in the list	
	count() - No of elements with specified value	02
	PoP() - Removes and returns the item	02

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Question Number	Solution	Marks Allocated
(c)	n = int (input ('Enter the no of element'))	
	num_list = [ ]	02
	for i in range(n) :	
	num = int(input('Enter no {}:'.format(i+1)))	
	num_list. append (num)	02
	mean = sum (sum_list) / n.	
	var = sum ([((x - mean) ** 2) for x in num_list]) / n	
	std = var ** 0.5	02
	Print ('mean', mean) Print('Variance', var)	
	Print ('std deviation', std)	
4 (a)	Tuples are like strings which are immutable.	
	Tuples are typed in parenthesis ( ) instead of square bracket in list.	02
		02
	Code optimization can be done on tuple	02
	Any other differences	
(b)	get( ) - returns the value of the item	
	items( ) - returns a view object	02
	keys( ) - returns keys in a dictionary	
	values( ) - returns values in a dictionary	02
	Eg :- spam = {'color' : 'red', 'age' : 22}	
	for v in spam. values( ):	02
	Print (v)	
	for k in spam. keys( ):	
	Print(k)	02
	for i in spam. items( ) :	
	Print (i)	

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Question Number	Solution	Marks Allocated
4(c)	a = int(input('Enter First No')) b = int(input('Enter second No')) a, b = b, a Print('The value of a after swapping', a) Print('The value of b after swapping', b)	
	MODULE - III	
5(a)	~~hello~~ (i) True (ii) 'hello' (iii) There_can_be_only_one	2×3 = 06
(b)	(i) isalpha() (ii) isalnum() (iii) isspace() (iv) startswith() (v) Join and split method ANY FIVE METHODS WITH Examples.	06

(c)	user_string = input('Enter a string')
	vowels = 0 consonants = 0
	for each_character in user_string:
	if (each_character == 'a' or each_character == 'e' or each_character == 'i' or 'o' or 'u')
	vowels += 1
	elif 'a' < each_character < 'z':
	consonants += 1
	Print('Total no of vowels in string {vowels}')
	Print('Total No. of consonants in string {consonants}')
		08

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Question Number	Solution	Marks Allocated
6 (a)	There are three steps to read or write Python files	02
	(1) call the open( ) function (2) call the read( ) or write( ) method on the file	02
	(3) close the file by calling close( ) method. Explain with Example.	03
(b)	There are two ways to specify a file	02
	(1) An absolute path always begins with root folder	02
	(2) Relative path, which is relative to the program's current working directory. Absolute C:\\ ├ Folder1 └ Folder2 : Relative .\\ ..\\ :	03
(c)	we can save Python programs to binary shelf file using shelve module. This way our programs can restore data to variables from hard drive.	04
	import shelve Explain with an Example	02
	MODULE-IV	
7 (a)	shutil module provides functions for copying file and folders.	02
	shutil. copy (source, destination)	02
	shutil. move (source, destination)	
	for deleting os. unlink (path) os. rmdir (path) with Example.	02

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Question Number	Solution	Marks Allocated
(b)	compressing a file reduces its size which is useful when transferring it over internet. Import Zipfile module	04
	Explain the reading of zipfile with Examples.	02
(c)	Shutil.copy() copies single file	02
	Shutil.copytree() copies Entire folder.	02
	Explain with Example	02
8(a)	Assertions is a sanity check for code	02
	Raise expection raises an exception whenever it tries to excute invalid code.	02
	Explain with syntax	02
(b)	def DivExp(a,b): assert a>0, 'value must be greater than zero'	03
	if b==0: raise Exception('Denominator Can't be zero')	02
	c = a/b return c	02
	a = int(input('Enter a value of a')) b = int(input('Enter a value of b')) Print('Divison of c = a/b = ', DivExp(a,b))	01
(c)	(i) DEBUG (ii) INFO (iii) WARNING (iv) ERROR (v) CRITICAL	04
	Explain briefly	02

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Question Number	Solution	Marks Allocated
	MODULE - V	
9 (a)	Class is a template that can be used to construct object Object is an instance of class	02
	class rectangle: height = 100.00 width = 200.00	02

	class point:
	x=0 y=0
	def find center (rect):
	c = point ()
	c.x = rect. corner.x + (rect. height/2)
	c.y = rect. corner.y + (rect. width/2)
		02

	return (c) rect. corner = point() corner = find centre(rect) print ('centre of rectangle (%g,%g)' % (center.x, center.y))	02
(b)	Prototyping and Planning two approaches to software development Explain briefly	06
(c)	__init () __ method is called when an instance of class is created __str() __ is called when an instance of class is converted into a string representation.	
10 (a)	one can change the state of object by making changes to one of its attributes. Explain with Eg	06

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Question Number	Solution	Marks Allocated
(b)	Polymorphism is the ability of objects to take on different forms (1) Overloading (2) Overriding Explain with Examples	02 03+03
(c)	Modifiers These methods modify the attributes of an object. Also called mutator or setter method	03
	Pure functions These function returns same o/p for an given i/p and has NO side effects. Explain with Examples.	03.